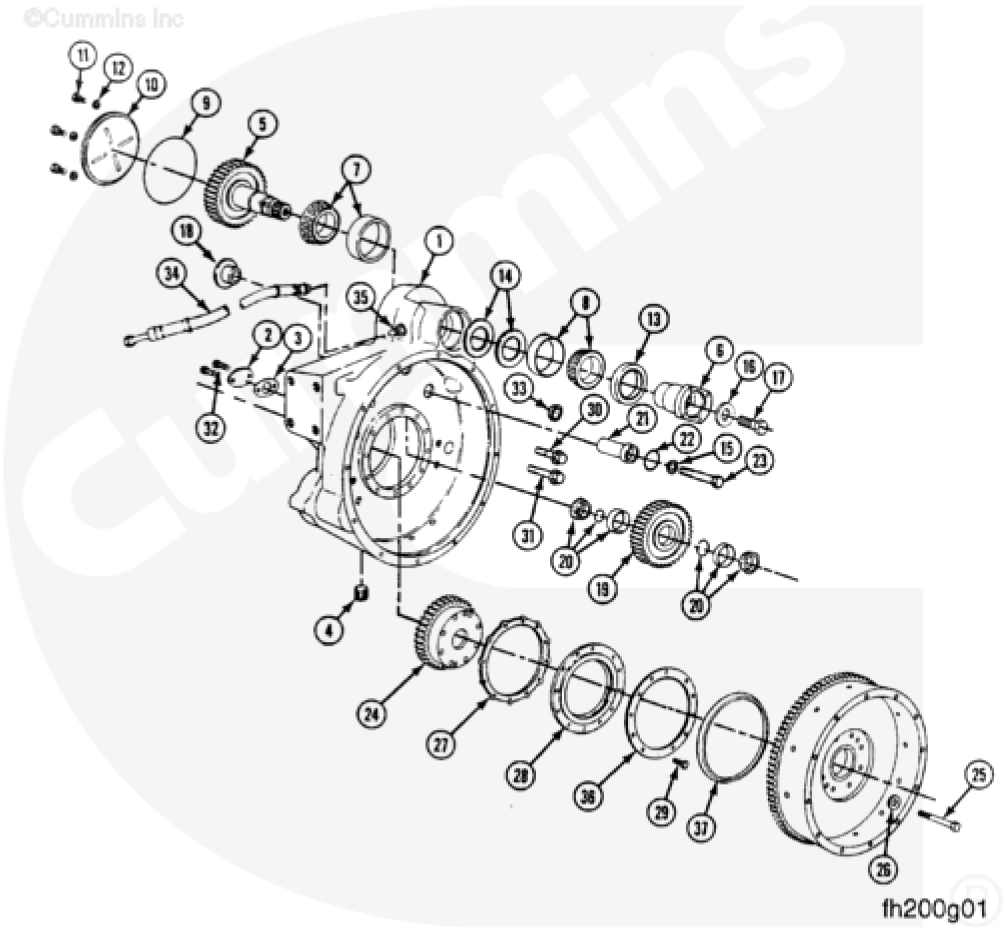
## Trainings

## Exploded View




1. Housing, Flywheel - 1
2. Cover, Access Hole - 1
3. Gasket, Cover Plate - 1
4. Plug, Threaded Drain - 1
5. Shaft, Accessory Drive Output - 1
6. Flange, Power Takeoff - 1
7. Bearing, Roller (Large) - 1
8. Bearing, Roller (Medium) - 1
9. Seal, Rectangular Ring (Tetra) - 1
10. Plate, Cover (Ribbed) - 1
11. Capscrew - 4
12. Washer, Plain - 4

13. Seal, Oil (Output Shaft) - 1
14. Shims (See next page)
15. Washer, Plain - 1
16. Washer, Plain (PTO Flange) - 1
17. Screw, Hexagon Head Cap - 1
18. Retainer, Shaft - 1
19. Gear, Idler - 1
20. Assembly, Bearing and Race - 1
21. Shaft, Idler - 1
22. Seal, O-ring - 2
23. Screw, Hexagon Head Cap - 1
24. Gear, Crankshaft - 1
25. Screw, Hexagon Head Cap - 8
26. Washer, Plain - 8
27. Gasket, Carrier - 1
28. Kit, Seal - 1
29. Screw, Captive Washer Cap - 12
30. Screw, Hexagon Head Cap - 5
31. Screw, Hexagon Head Cap - 7
32. Screw, Hexagon Head Cap - 2
33. Plug, Expansion - 1
34. Hose, Flexible Oil Supply - 1
35. Connection, Mate Oil Supply - 1
36. Ring, Clamping - 1
37. Seal, Dust - 1

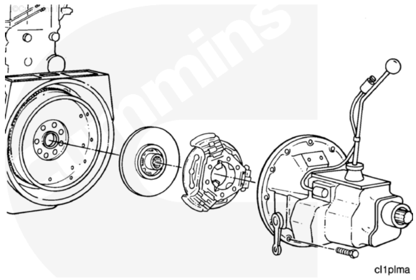
**There are 7 shims available. A given REPTO could have any combination of these shims.**

| Ref. No. | Description | mm | in |
|---|---|---|---|
| 14 | Shim | 0.127 | 0.005 |
| 14 | Shim | 0.254 | 0.010 |
| 14 | Shim | 0.381 | 0.015 |
| 14 | Shim | 0.051 | 0.002 |
| 14 | Shim | 0.076 | 0.003 |
| 14 | Shim | 0.508 | 0.020 |
| 14 | Shim | 1.016 | 0.040 |

## Remove

**⚠ WARNING ⚠**

**This assembly weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift the assembly.**

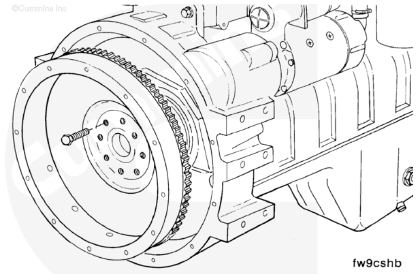



Remove the transmission, clutch, and related components, if equipped.

Refer to the OEM's instructions.

**⚠ WARNING ⚠**

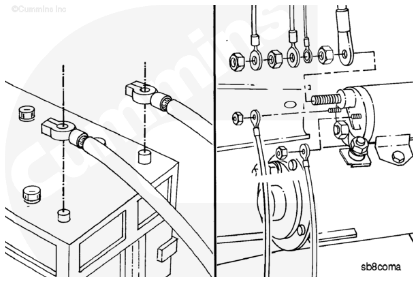
**This assembly weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift the assembly.**




Remove the flywheel. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/100/100-016-005.html)

**⚠ WARNING ⚠**

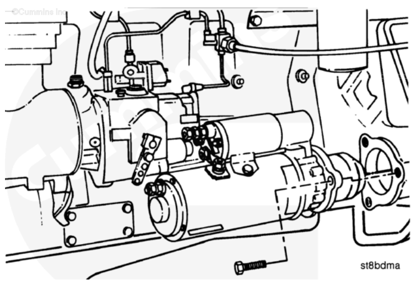
**Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arching, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**




Disconnect the electrical connections from the battery.

Tag and disconnect the electrical connections from the starting motor.

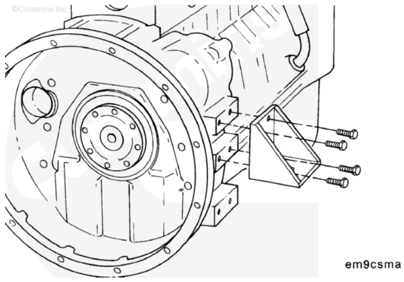
Remove the starter; refer to Refer to Procedure 013-020 in Section 13. (/qs3/pubsys2/xml/en/procedures/100/100-013-020.html)



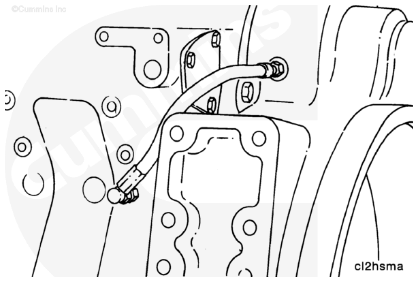

**⚠ WARNING ⚠**

**Use a floor jack or other suitable lifting fixture to support the engine. Personal injury can result.**

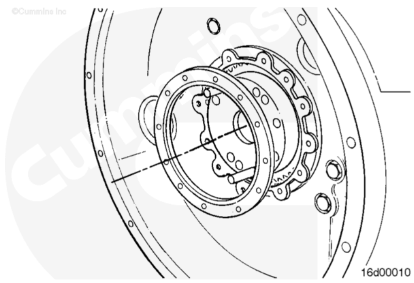
Remove the capscrews and both rear engine mounts.




Remove the REPTO oil supply line.



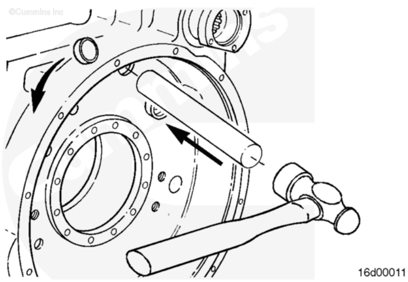

Remove the oil seal capscrews, the oil seal, and the gasket.




To gain access to the housing capscrews, use a drift to drive the cup plugs straight through into the housing.

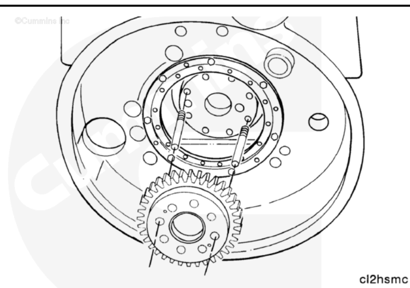
**Note : Do not attempt to back the plugs out or rotate the plugs out. The cup plug bore will be damaged and oil leakage will occur.**

Retrieve the plugs from inside the housing.

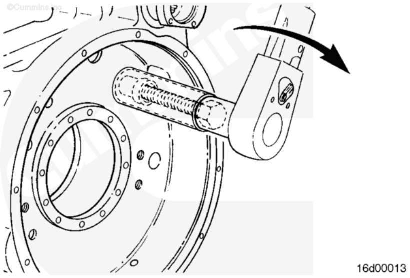



Install two crankshaft locator studs, Part Number 3822784, into the crankshaft flywheel mounting flange 180 degrees apart.

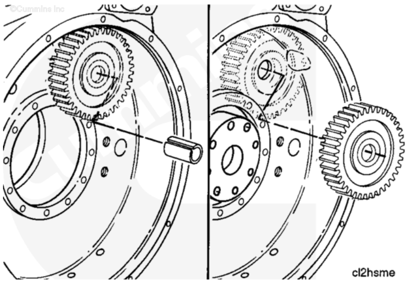
Remove the crankshaft drive gear.




Use the idler shaft puller, Part Number 3823709, to remove the idler shaft.

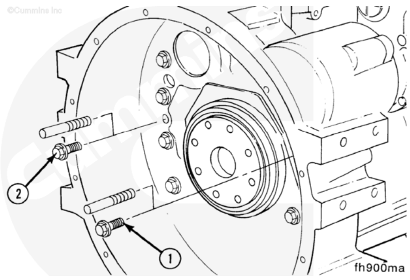


Remove the idler gear to gain access to the rest of the housing capscrews.



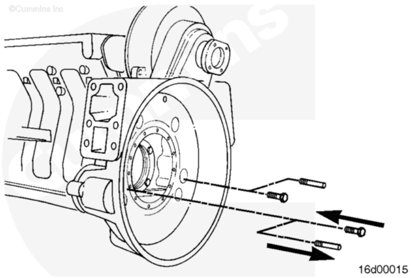
Remove two of the capscrews and install two guide pins, Part Number 3376638, to support the housing during removal.

Use offset wrench, Part Number 3823892, to remove the capscrews which are **not** in view.




**⚠ WARNING ⚠**

**This assembly weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift the assembly.**



Remove the remaining capscrews. Use a rubber hammer to loosen the housing.

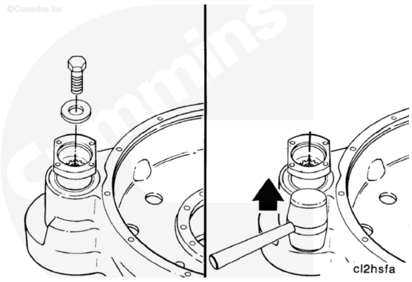
Remove the housing.

# Disassemble

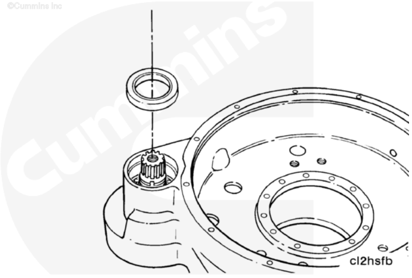
**Note : Use gear locking tool, Part Number 3823891, to prevent the output shaft from turning when removing the retainer capscrew.**

Remove the capscrew and washer that secures the output flange to the output shaft.

Use a rawhide hammer to remove the output flange and flat washer from the output shaft.




Use a dent puller to remove the seal. Do **not** damage the surface of the housing or seal bore.

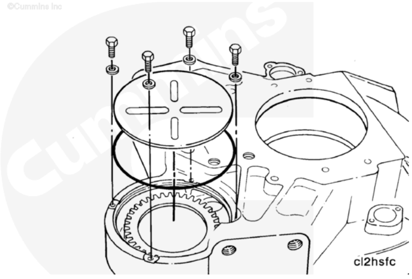



**Note : When the housing is turned over, the bearing will fall out. Do not allow the bearing to be damaged.**

Turn the housing over so the four cover plate capscres are accessible. Be careful **not** to damage the output shaft.

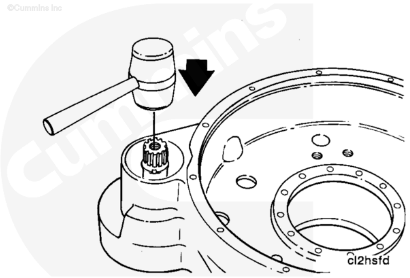
Remove the capscrews and ribbed cover plate from the output gear housing.

Remove and discard the square cut o-ring seal.

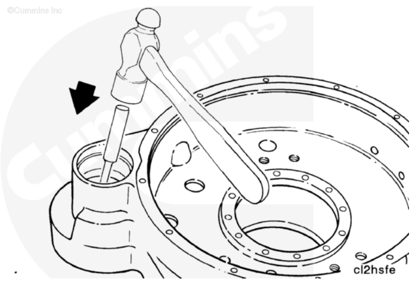



**Note : Save the original shims for rebuild purposes. They will be used to set the proper end clearance on the output shaft and bearing assembly.**

Turn the housing over and use a rawhide hammer to hit the end of the output shaft to remove the poutput shaft subassembly from the REPTO housing.



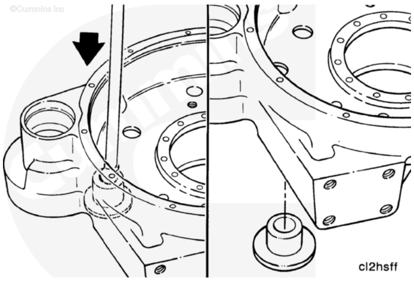

With the housing positioned so the cylinder block mating surface is down, use a hammer and brass punch to drive the bearing outer races out of the output shaft housing bore.



Support the housing in a press with the cylinder block mating surface down.

With a long mandrel, press out the idler shaft bushing.

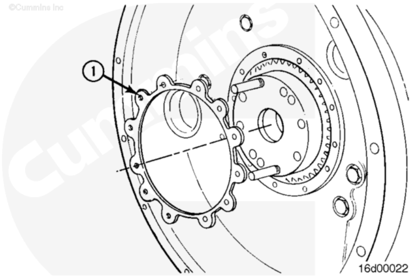
Remove and discard o-ring.




**Note : Save the original shims for rebuild purposes, to set the proper end clearance on the output shaft and bearing assembly.**

Use a hammer and brass drift to remove the two bearing outer races from the bore of the idler gear. Discard the outer races.

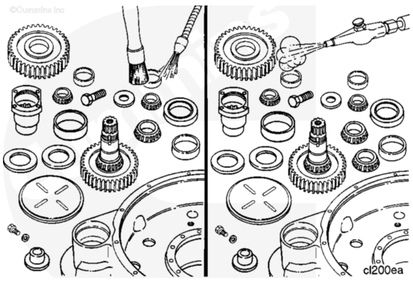
Remove the large spacer ring from the center groove of the gear. Discard the spacer ring.




## Clean

**⚠ WARNING ⚠**

**When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.**



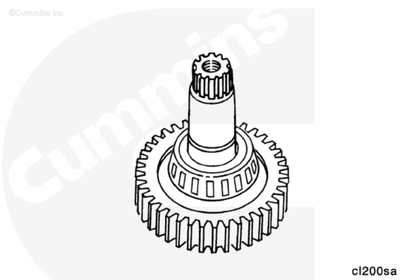

Use a steam cleaner to clean all areas of the idler gear.

## Inspect for Reuse

Inspect the output shaft and bearings for wear.

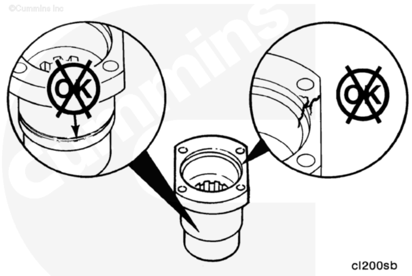
Inspect the output gear for damage.

Replace if defective.



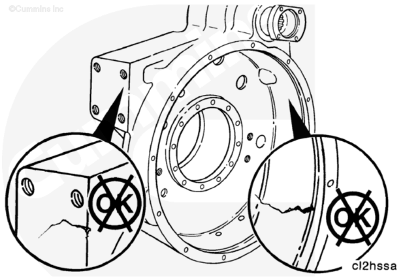
Inspect the output flange for damage or wear grooves from the oil seal.

Replace if defective.



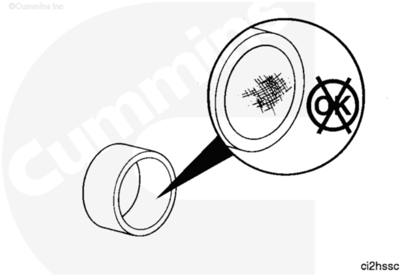

Inspect the REPTO housing for cracks at the rear engine mounting surfaces and the flywheel bore.

Replace the housing if cracked.



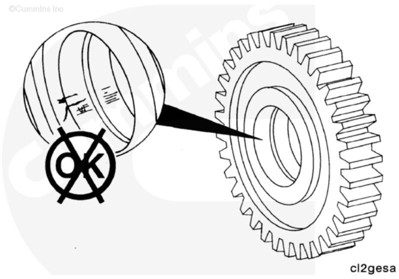

Inspect the idler shaft bushing for wear.

Replace the bushing if worn.




Closely inspect the bore, the side faces and the teeth of the idler gear.

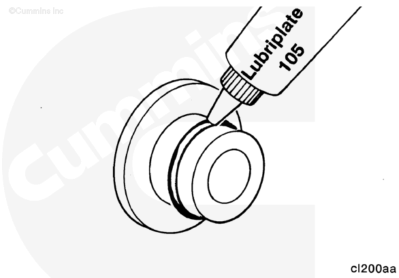
Replace the gear if there are cracks, discoloration from excessive heat or other damage.




## Assemble

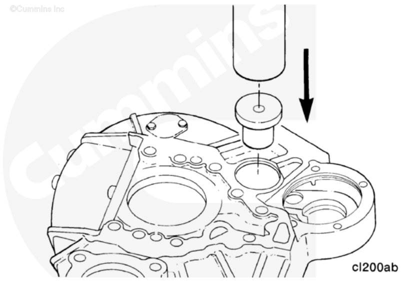
Install a new o-ring on the idler shaft bushing.

Use Lubriplate™ 105 to lubricate the o-ring.



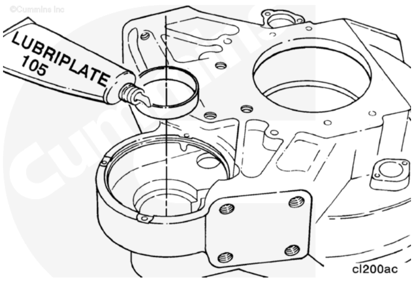
Support the housing evenly with the engine mating surface up.

Press the new bushing into the housing until it is below the surface of the cylinder block mating surface.




Use Lubriplate™ 105 to lubricate the larger bearing race.

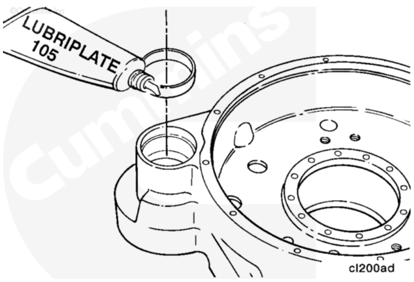
Use the larger end of driver, Part Number 3823893, to press the bearing race to the shoulder in the housing.




Turn the housing over and support evenly in the press.

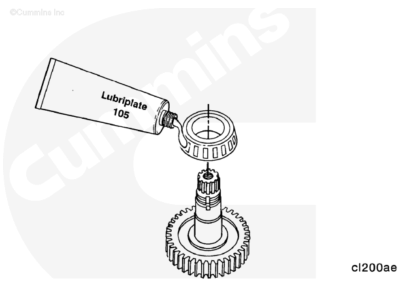
Use Lubriplate™ 105 to lubricate the smaller bearing race.

Use the smaller end of driver, Part Number 3823893, to press the bearing race to the shoulder in the housing.



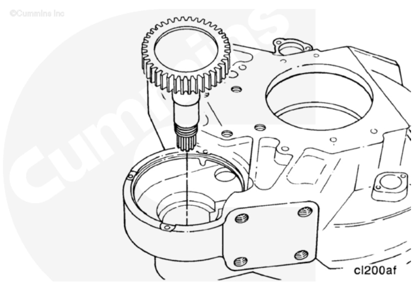

Use Lubriplate™ 105 to lubricate the output shaft and larger bearing.

Install the larger bearing onto the output shaft.




Position the housing on the table so the cylinder block mating surface is up.

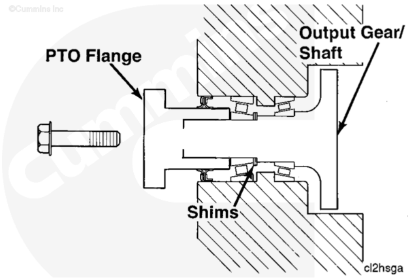
Install the output shaft assembly into the housing.



The correct end clearance (rolling resistance) is determined by the number and thickness of shims used between the two bearings.

When the shim thickness is increased, there is more shaft end clearance and less rolling resistance.

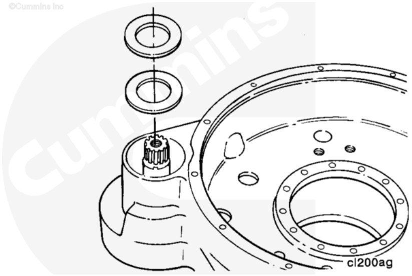
When the shim thickness is decreased, there is less shaft end clearance and more rolling resistance.




Turn the housing over with engine mating surface down, while holding the output shaft and gear in place.

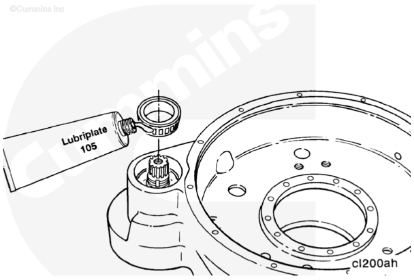
**Note : If the original thickness of shims is not available for reuse, begining thickness of 1.47 mm [0.058 in] can be used as a starting point.**

Install the original thickness of shims.



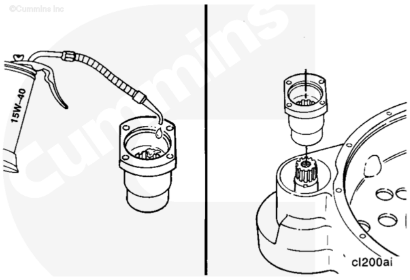

Use Lubriplate™ 105 to lubricate the smaller bearing.

Install the smaller bearing onto the shaft.



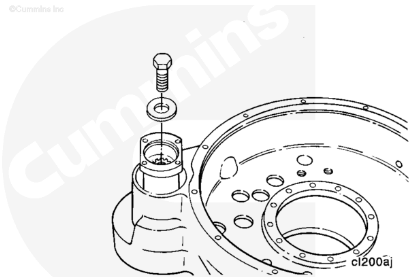

Temporarily install the output flange, before installing the oil seal.

Use clean 15W-40 oil to lubricate the splines.



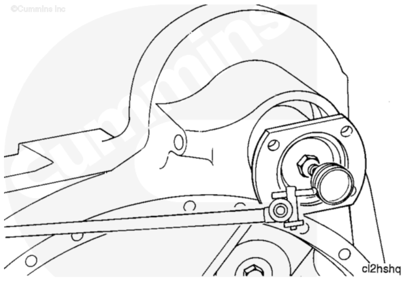

Install the flat washer and capscrew. Use the gear locking tool, Part Numnber 3823891, to hold the output shaft while tightening the capscrew.

**Torque Value:** 205 n•m [ 150 ft-lb ]



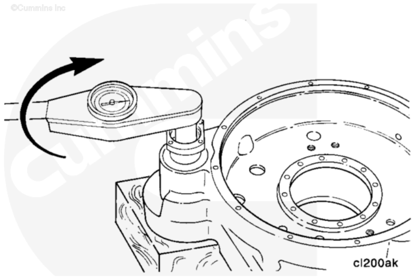

Measure shaft end play.

| Output Shaft End Play | | |
|---|---|---|
| mm | | in |
| 0.00 | MIN | 0.000 |
| 0.03 | MAX | 0.001 |




Check the output shaft rolling resistance with a torque wrench.

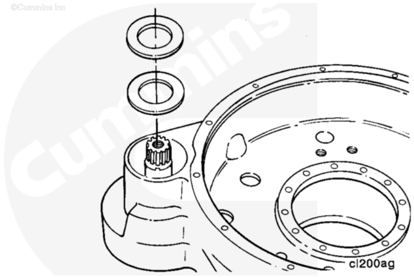
Rolling resistance **must**be between 0.6 to 1 N•m [5 to 10 in-lb].




If the rolloing resistance is **not** within specification, remove the output flange and smaller bearing.

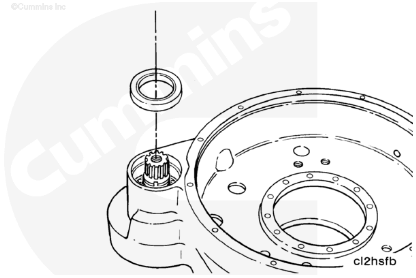
Add or subtract shims to obtain the correct rolling resistance.

**Note : Adding more shims will decrease the resistance and removing shims will increase resistance. Any combination of shims can be used.**




Once the correct rolling resistance is obtained, remove the output flange and install a new oil seal.

Press the oil seal flush with the housing surface.

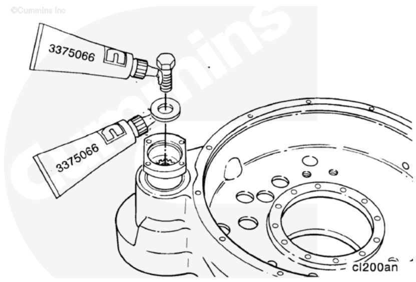



Apply pipe sealant, Part Number 3375066, to the output flange capscrew and under the washer.

Install the output flange flat washer and capscrew.

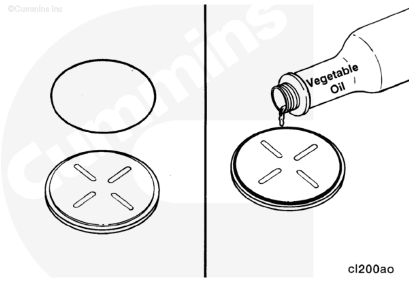
Tighten the capscrew.

**Torque Value:** 205 n•m [ 150 ft-lb ]



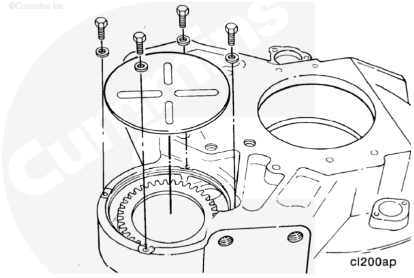

Install a new o-ring on the bearing housing cover.

Use clean vegetable oil to lubricate the o-ring.



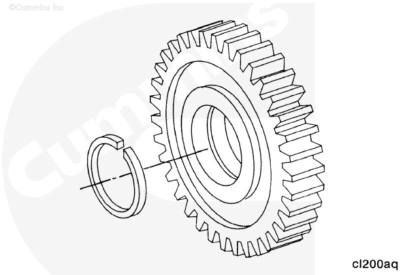

Install and tighten the cover and four capscrews.

**Torque Value:** 18 n•m [ 14 ft-lb ]




Insert a new spacer ring into the bore of the idler gear.

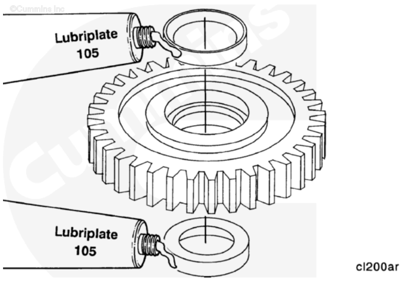
Push it in until it snaps into place in the center groove.




Use Lubriplate™ 105 or equivalent to lubricate the bearing outer races.

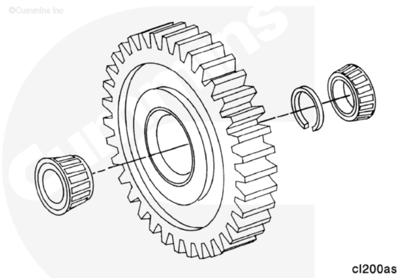
Press the two new bearing outer races into the bore of the idler gear.

The larger side of the taper **must** face toward the outside of the gear.



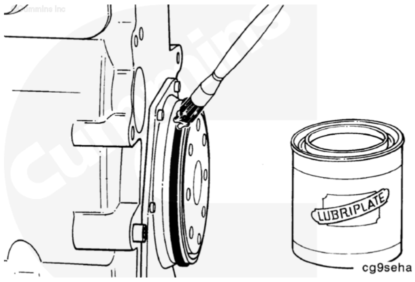

**Note : Do not interchange individual parts that make up the idler gear bearing assembly. Rebuild the idler gear with bearings that are packaged together.**

Keep the two roller bearing assemblies and the spacer ring with the idler gear.

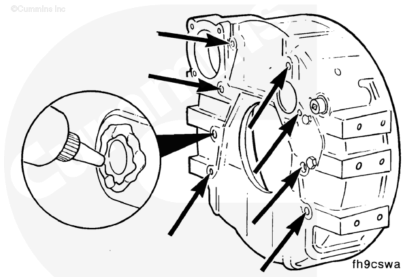


# Install

Install rectangular seal and lubricate with Lubriplate™ 105, or equivalent.

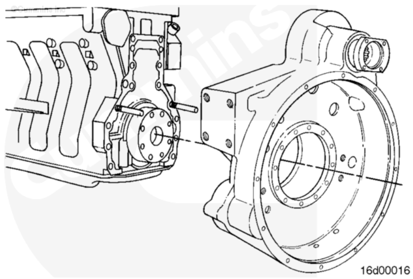



Apply a continuous bead of ThreeBond™, or equivalent, around all capscrew holes on the mounting surface of the flywheel housing.




**⚠ WARNING ⚠**

**The component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift the component.**



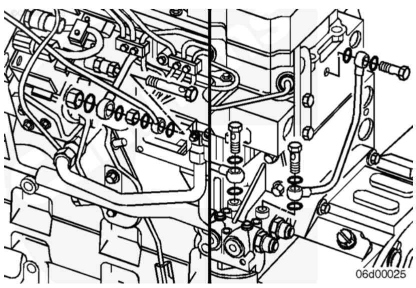

Inspect the rear face of the cylinder block and flywheel housing mounting surface for cleanliness and nicks or burrs.

Install the flywheel housing over the two ring dowels.

**Note : Be sure the sealing ring is not damaged during installation.**

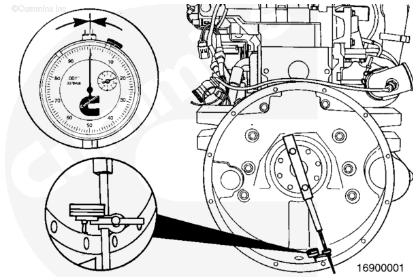
Install the capscrews and tighten with the sequence shown, using offset wrench, Part Number 3823892, for capscrews hidden from view.

**Torque Value:** 60 n•m [ 45 ft-lb ]




**Note : Before installing the idler gear, measure the flywheel housing bore and face alignment. Refer to Procedure 016-006.**

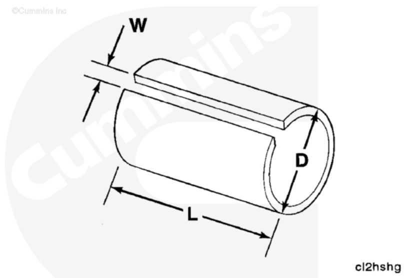
**(/qs3/pubsys2/xml/en/procedures/100/100-016-006.html)**




Fabricate a sleeve from 38.10 mm [1.50 in] PVC (or equivalent) to the following dimensions.

Length - 25.4 mm [1.0 in]

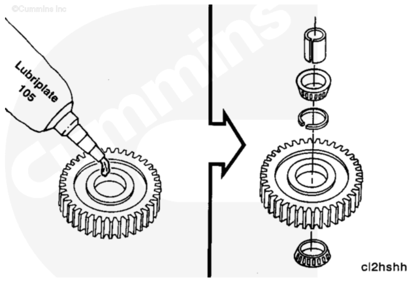
Slot - 12.70 mm [0.50 in]




**Note : The outer bearing races of new replacement gears are already pressed into the gear.**

Apply a thin film of Lubriplate™ 105 or equivalent on the outer races and the bearings.

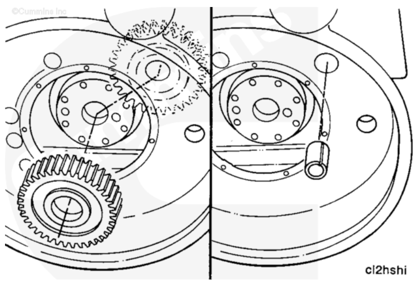
Install the bearing and spacer into the idler gear. Use a plastic sleeve to hold the bearing assembly together when installing the idler gear assembly.




Apply a thin film of Lubriplate™ 105 or equivalent into the idler shaft bore of the housing and on the idler shaft.

Install the idler gear assembly into the flywheel housing.

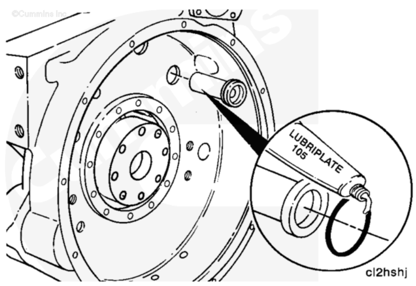
Hold the idler gear and bearing in place and remove the plastic sleeve.




Use clean Lubriplate™ 105 to lubricate the idle shaft o-ring and install the o-ring into the shaft.

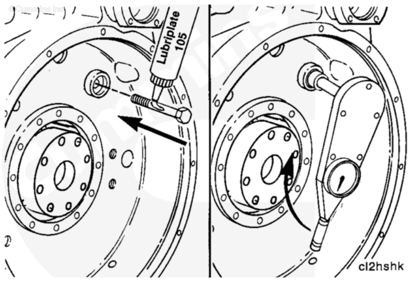
Hold the gear assembly in place and insert the idler shaft through the housing and idler gear bearings.

**Note : Do not use a hammer when installing the idler shaft and capscrew or the part can be damaged.**



Apply Lubriplate™ 105 under the head of the idler shaft capscrew. Insert the capscrew through the idler shaft. Tighten the installation capscrew with a torque wrench.

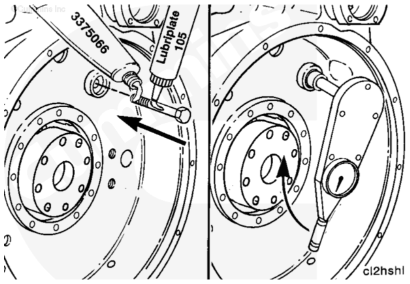
**Note : The toque needed to draw the idler shaft in place must not exceed 88 N•m [65 ft-lb]. If installation torque exceeds this amount, it is an indication of misalignment between the bore and the shaft. Remove the idler shaft and install it again.**




When the idler shaft has been seated, remove the capscrew.

Apply pipe sealant, Part Number 3375066, to the threads of the idler shaft capscrew. Apply Lubriplate™ 105 under the head of the capscrew and tighten to it's final torque value.

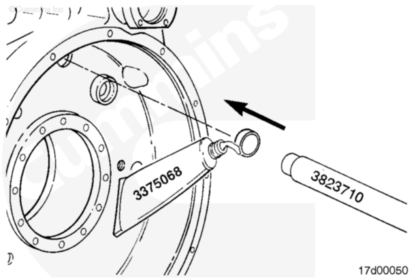
**Torque Value:** 105 n•m [ 75 ft-lb ]



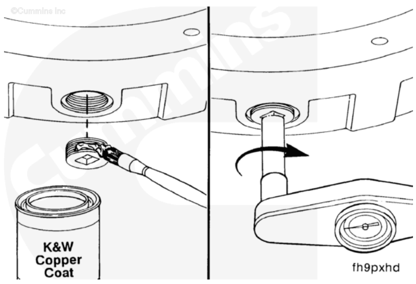

Apply a film of cup plug sealant, Part Number 3375068, to the outside diameter of the cup plugs.

Use a driver, Part Number 3823710, to install the cup plugs into the housing as shown.

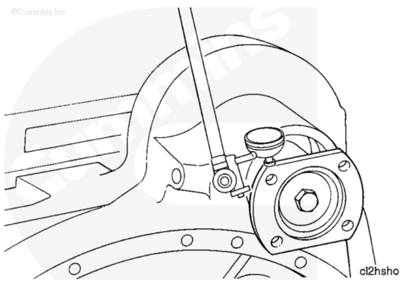
**Note : When installing cup plugs, make sure they are flush with the spot face on the flywheel housing and are not crooked.**




- Coat the flywheel housing drain plug with pipe sealant and install in the hole in the bottom of the flywheel housing.
- Tighten the plug.
- Refer to the pipe plug torque values in section 017 for different plug sizes.



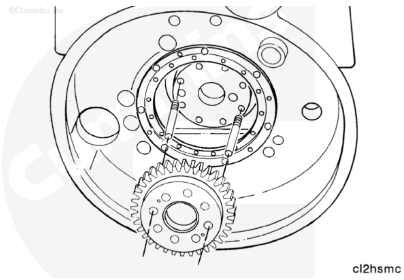

Turn the output flange so that the flat sides are on the top and bottom. This prevents any interference when the transmission is installed onto the housing.



Install two crankshaft locator studs, Part Number 3822784, into the crankshaft flywheel mounting flange 180 degrees apart.

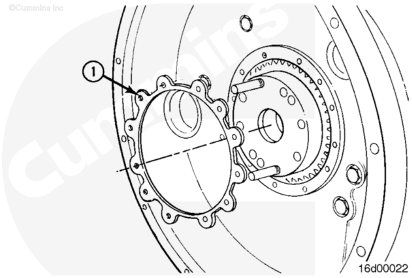
Make sure the crankshaft and crankshaft gear is clean.

Install the crankshaft gear in the locator studs.

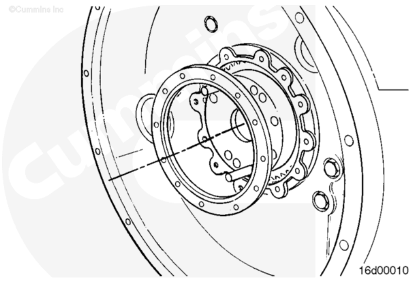



**Note : Do not use any kind of lubricant to install the seal. The oil seal must be installed with the crankshaft gear seal contact surface and the lip of the seal clean and dry to provide a proper oil sealing surface.**

Install a new gasket (1) in the flywheel housing.




Install a new seal over the crankshaft gear seal contact surface.

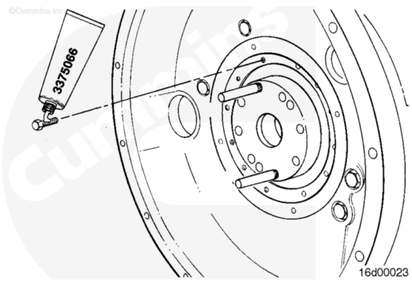



Apply sealant, Part Number 3375066, to seal retainer capscrews.

Install the capscrews, and tighten in a star pattern.

**Torque Value:**

1. 7 n•m [ 60 in-lb ]
2. 19 n•m [ 170 in-lb ]




**Last Modified: 26-Jun-2009**